

1. Hypothesis

After NIFTY 50 falls 2% or more close-to-close in one day, the average 5-day forward return is higher than the average 5-day forward return on normal (non-event) days.

2. Event definition

An event is a day with a close-to-close return of -2% or worse. I chose -2% because it is a genuine shock, but it still happens often enough to give a usable sample (145 raw events in 2007-2017). For robustness I also test -1.5% and -3%.

3. Recovery definition

Recovery means the forward return after the event is higher than the average forward return on normal days. I also report the share of trades with a positive return (win rate).

4. Entry and exit

The size of the fall is only known at the close, so buying at that close would use information I did not have. Entry is therefore the next trading day's open. Exit is the close of day 5, counting the entry day as day 1.

5. Holding period

The main test uses 5 trading days, fixed before looking at any results. Robustness holds: 1, 3 and 10 days.

6. Overlapping events

Falls cluster in time (2008, 2020), so nearby events are not independent. A new event is ignored while a previous trade is still open. This reduces 145 raw events to 92 in the development period.

7. Baseline

All non-event days over the same period, using the same entry and exit rule.

8. Test period

Development period: September 2007 to December 2017. Out-of-sample period: January 2018 onward, kept unseen until the primary test and robustness grid were finished, and then run once. Trades that would end after a period boundary are dropped, so development results never use out-of-sample prices.

9. Transaction costs

0.05% per side for brokerage and slippage, so 0.10% round trip.

10. Key assumptions

- Data is Yahoo Finance ^NSEI (daily, price index). Dividends are ignored
- Trades execute at the quoted open and close. Real fills in a crash would be worse
- Event returns are resampled with a standard bootstrap. Baseline returns overlap, so they are resampled in 20-day blocks
- One primary test (-2%, 5 days) was fixed in advance. The other 11 cells of the grid are only a stability check, judged against a Bonferroni cutoff of $0.05/12 = 0.0042$
- One position at a time. The sample includes the 2008 and 2020 crashes

11. Result

Development (n = 92): event mean 0.07% versus baseline 0.118%. The difference is, 0.05%, with a 95% CI of about [-1.0%, +0.9%].

Out-of-sample (n = 39): event mean -0.107% versus baseline 0.116%. The difference is, 0.22%, with a 95% CI of [-1.81%, +1.11%].

Robustness grid: all 12 confidence intervals include zero. The smallest p-value is 0.246, far above the Bonferroni cutoff. The signs are mixed, and hold = 1 is negative at every threshold.

Net of costs, the average trade lost money (-0.03% in development, -0.21% out-of-sample). The simple backtest (131 trades) lost 22.7% net.

The hypothesis is not supported.

12. What would change my mind

A difference with the same sign across neighbouring thresholds and holding periods, a confidence interval excluding zero in both periods, and a positive net mean after costs. None of these were observed.

13. Limitations

Small samples give wide confidence intervals (about +-1 to 2%), so a small real effect could not be detected. Events cluster in 2008 and 2020. Yahoo is a single unaudited source, and one date (2026-09-22) is missing and unexplained. Excluding 2008 and 2020 flips the difference to +0.57% (CI [-0.16, 1.31]), but this was found after seeing the results, so it is exploratory only.